

STAT 6410

Homework 1

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1. A.3 Let $I = [0, 1]$, $\Omega = \mathbb{R}$ and for $\alpha \in \mathbb{R}$, $A_\alpha = (\alpha - 1, \alpha + 1)$, the open interval $\{x : \alpha - 1 < x < \alpha + 1\}$.

- (a) Show that $\cup_{\alpha \in I} A_\alpha = (-1, 2)$ and $\cap_{\alpha \in I} A_\alpha = (0, 1)$.
(b) Suppose $J = \{x : x \in I, x \text{ is rational}\}$. Find $\cup_{x \in J} A_x$ and $\cap_{x \in J} A_x$.

Ans: (a)

Proof. $A_0 = (-1, 1)$ and $A_1 = (0, 2)$. Thus, $A_0 \cup A_1 = (-1, 2)$.

Part 1: WTS $\cup_{\alpha \in I} A_\alpha \subset A_0 \cup A_1$.

Let $x \in \cup_{\alpha \in I} A_\alpha$ be given.

Thus, $\exists \beta \in [0, 1]$ s.t. $x \in A_\beta$.

If this $\beta = 1$, $x \in A_1$.

If this $\beta \neq 1$: Since $\beta - 1$ can be at most 0 and $\beta + 1$ can be at least 1, we know that $0 \in A_\beta$. If $x \in (0, \beta + 1)$, $x \in (0, 2) = A_1$ because $(0, \beta + 1) \subset (0, 2)$. If $x \in (\beta - 1, 0)$, $x \in (-1, 1) = A_0$ because $(\beta - 1, 0) \subset (-1, 1)$.

Thus, $x \in A_0 \cup A_1$.

$\therefore \cup_{\alpha \in I} A_\alpha \subset A_0 \cup A_1$

Part 2: WTS $A_0 \cup A_1 \subset \cup_{\alpha \in I} A_\alpha$.

This is trivially true

Thus, $\cup_{\alpha \in I} A_\alpha = A_0 \cup A_1 = (-1, 2)$

Proof. $A_0 = (-1, 1)$ and $A_1 = (0, 2)$. Thus, $A_0 \cap A_1 = (0, 1)$.

Part 1: WTS $\cap_{\alpha \in I} A_\alpha \subset A_0 \cap A_1$

Let $x \in \cap_{\alpha \in I} A_\alpha$ be given.

Since, $\max_{\alpha \in I}(\inf(A_\alpha)) = \inf(A_1) = 0$ and similarly, $\min_{\alpha \in I}(\sup(A_\alpha)) = \sup(A_0) = 1$.

Thus, $0 \leq x \leq 1$. Therefore, $x \in (0, 1) = A_0 \cap A_1$

Part 2: WTS $A_0 \cap A_1 \subset \cap_{\alpha \in I} A_\alpha$

This is trivially true.

Thus, $\cap_{\alpha \in I} A_\alpha = A_0 \cap A_1 = (0, 1)$

(b)

Part 1: Since $\mathbb{Q} \subset \mathbb{R}$, it is clear to see that $\cup_{x \in J} A_x \subset \cup_{x \in I} A_x$.

We know that rational numbers are dense on real numbers. This means that $\forall \alpha \in I, \exists p, q \in \mathbb{Q} \cap I$ s.t. $p < \alpha < q$.

So, $A_\alpha = (\alpha - 1, \alpha + 1) \subset (p - 1, q + 1) = (p - 1, p + 1) \cup (q - 1, q + 1) = A_p \cup A_q \subset \cup_{x \in J} A_x$.

Therefore, $\cup_{x \in J} A_x = \cup_{\alpha \in I} A_\alpha = (-1, 2)$

Part 2: Since, $\mathbb{Q} \subset \mathbb{R}$, it is clear to see that $\cap_{\alpha \in I} A_\alpha \subset \cap_{x \in J} A_x$.

$\cap_{x \in J} A_x \subset A_0 \cap A_1 = (0, 1) = \cap_{\alpha \in I} A_\alpha$.

Thus, $\cap_{x \in J} A_x = \cap_{\alpha \in I} A_\alpha = (0, 1)$

□

□

2. A.5 Show that $X \equiv \{0, 1\}^{\mathbb{N}}$, the set of all sequences $\{\omega_i\}_{i \in \mathbb{N}}$ where each $\omega_i \in \{0, 1\}$, is *uncountable*. Conclude that $\mathcal{P}(\mathbb{N})$ is uncountable.

Ans:

Proof. $(0, 1)$ is an uncountable set.

If X is uncountable, there must be a one-to-one function, $f : (0, 1) \rightarrow X = \{0, 1\}^{\mathbb{N}}$, that maps from $(0, 1)$ to X .

Suppose we think of elements, x , in X as digits of a base 2 number. For instance, $x = (1, 1, 0, 0, \dots)$ are the 1st, 2nd, 3rd, ... digits of number in base 2. ...0000011. This would be 3 in base 10.

Here is a verbal description of $f : (0, 1) \rightarrow X$:

- i. Start with an element of $(0, 1)$. Let's call it a .
- ii. Convert to a natural number, a^* , where the 10th place in a is the ones place in a^* , the 100th place in a is the tens place in a^* , the 1000th place in a is the hundreds place in a^* and so forth. For instance, if $a = 0.9200000\dots$, $a^* = 29$
- iii. Convert to a^* to base 2. For instance if $a^* = 29$, it is ...00000010100 in base 10.

- iv. Now, we can bring this into $\{0, 1\}^{\mathbb{N}}$, where the first digit of the binary version of a^* is the first element in the vector in X , the second digit in the binary version of a^* is the 2nd element in the vector of X , and so forth. For example, 10100 would be $\{0, 0, 1, 0, 1, 0, 0, \dots\}$.

Conversion from base 10 to base 2 is inherently one-to-one and onto. Each element in $a \in (0, 1)$ can uniquely be transformed to a natural number (base 10) as described in (ii), and each natural number can be uniquely transformed to a base 2 number as described in (iii), which can be vectorized uniquely as described in (vi).

Thus, $f : (0, 1) \rightarrow X$ is one-to-one.

Thus, $|X| = |(0, 1)|$, and X is uncountable.

Now we can define a one-to-one (and onto) mapping between them, $g : \mathcal{P}(\mathbb{N}) \rightarrow \{0, 1\}^{\mathbb{N}}$:

For all $Z \in \mathcal{P}(\mathbb{N})$, $g(Z) = \{x_i\}_{i \in \mathbb{N}}$, where $x_i = \begin{cases} 1 & \text{if } i \in Z \\ 0 & \text{if } i \notin Z \end{cases}$

Thus, we can say that $|X| = |\mathcal{P}(\mathbb{N})|$. Thus $\mathcal{P}(\mathbb{N})$ is uncountable. $\square$

3. A.9 Apply the principle of induction to establish the following:

- (a) For each $n \in \mathbb{N}$, $\sum_{j=1}^n j^2 = \frac{n(n+1)(2n+1)}{6}$
- (b) For each $n \in \mathbb{N}$, $x_1, x_2, \dots, x_k \in \mathbb{R}$,
- (i) (*The binomial formula*). $(x_1 + x_2)^n = \sum_{r=0}^n \binom{n}{r} x_1^r x_2^{n-r}$
- (ii) (*The multinomial formula*). $(x_1 + x_2 + \dots + x_k)^n = \sum \frac{n!}{r_1! r_2! \dots r_k!} x_1^{r_1} x_2^{r_2} \dots x_k^{r_k}$,
where the summation extends over all $(r_1, r_2, \dots, r_k)$ such that $r_i \in \mathbb{N}$,
 $0 \leq r_i \leq n$, $\sum_{i=1}^k r_i = n$.

Ans: (a)

Proof. Let $\mathcal{P}(n)$ be the mathematical statement $\sum_{j=1}^n j^2 = \frac{n(n+1)(2n+1)}{6}$.

i) WTS $\mathcal{P}(1)$ is true.

Left Side: $\sum_{j=1}^1 j^2 = 1$

Right Side: $\frac{1(1+1)(2(1)+1)}{6} = \frac{6}{6} = 1$ Since the left side = right side, $\mathcal{P}(1)$ holds true.

ii) WTS that $\mathcal{P}(n)$ is true $\implies \mathcal{P}(n+1)$ is true.

Let $n \in \mathbb{N}$ be given. Suppose $\mathcal{P}(n)$ is true.

Thus, $\sum_{j=1}^n j^2 = \frac{n(n+1)(2n+1)}{6}$.

$$\begin{aligned}\sum_{j=1}^{n+1} j^2 &= \sum_{j=1}^n j^2 + (n+1)^2 = \frac{n(n+1)(2n+1)}{6} + (n+1)^2 = \frac{2n^3+3n^2+n}{6} + \frac{6n^2+12n+6}{6} \\ &= \frac{2n^3+9n^2+12n+6}{6} = \frac{(n+1)((n+1)+1)(2(n+1)+1)}{6}\end{aligned}$$

Thus, $\mathcal{P}(n+1)$ is true.

By induction, we can say that $\mathcal{P}(n)$ is true for all $n \in \mathbb{N}$ □

(b.i)

Proof. Let $\mathcal{P}(n)$ be the mathematical statement $(x_1 + x_2)^n = \sum_{r=0}^n \binom{n}{r} x_1^r x_2^{n-r}$.

i) WTS $\mathcal{P}(1)$ is true.

Left Side: $(x_1 + x_2)^1 = x_1 + x_2$

Right Side: $\sum_{r=0}^1 \binom{1}{r} x_1^r x_2^{1-r} = \binom{1}{0} x_1^0 x_2^{1-0} + \binom{1}{1} x_1^1 x_2^{1-1} = x_1 + x_2$

Since The LHS = RHS, $\mathcal{P}(1)$ is true.

ii) WTS that $\mathcal{P}(n)$ is true $\implies \mathcal{P}(n+1)$ is true.

Let $n \in \mathbb{N}$ be given. Suppose $\mathcal{P}(n)$ is true.

Thus, $(x_1 + x_2)^n = \sum_{r=0}^n \binom{n}{r} x_1^r x_2^{n-r}$

$$\begin{aligned}(x_1 + x_2)^{n+1} &= (x_1 + x_2)^n (x_1 + x_2) = \sum_{r=0}^n \binom{n}{r} x_1^r x_2^{n-r} (x_1 + x_2) \\ &= x_1 \sum_{r=0}^n \binom{n}{r} x_1^r x_2^{n-r} + x_2 \sum_{r=0}^n \binom{n}{r} x_1^r x_2^{n-r} = \sum_{r=0}^n \binom{n}{r} x_1^{r+1} x_2^{n-r} + \sum_{r=0}^n \binom{n}{r} x_1^r x_2^{n+1-r} \\ &= \binom{n}{n} x_1^{n+1} + \sum_{r=0}^{n-1} \binom{n}{r} x_1^{r+1} x_2^{n-r} + \binom{n}{0} x_2^{n+1} + \sum_{r=1}^n \binom{n}{r} x_1^r x_2^{n+1-r} \\ &= x_1^{n+1} + \sum_{r=1}^n \binom{n}{r-1} x_1^r x_2^{n+1-r} + x_2^{n+1} + \sum_{r=1}^n \binom{n}{r} x_1^r x_2^{n+1-r} \\ &= \sum_{r=0}^0 \binom{n+1}{r} x_1^r x_2^{n+1-r} + \sum_{r=n+1}^{n+1} \binom{n+1}{r} x_1^r x_2^{n+1-r} + \sum_{r=1}^n \binom{n+1}{r} x_1^r x_2^{n+1-r}\end{aligned}$$

By Pascal's Law

$$= \sum_{r=0}^{n+1} \binom{n+1}{r} x_1^r x_2^{n+1-r}$$

Thus, $\mathcal{P}(n)$ is true $\implies \mathcal{P}(n+1)$. Thus, by induction:

For each $n \in \mathbb{N}$, $x_1, x_2, \dots, x_k \in \mathbb{R}$,

$$(x_1 + x_2 + \dots + x_k)^n = \sum \frac{n!}{r_1! r_2! \dots r_k!} x_1^{r_1} x_2^{r_2} \dots x_k^{r_k},$$

where the summation extends over all $(r_1, r_2, \dots, r_k)$ such that $r_i \in \mathbb{N}$,

$$0 \leq r_i \leq n, \sum_{i=1}^k r_i = n.$$

□

(b.ii)

Proof. Let $\mathcal{P}(n)$ be the mathematical statement

$(x_1 + x_2 + \dots + x_k)^n = \sum \frac{n!}{r_1!r_2!\dots r_k!} x_1^{r_1} x_2^{r_2} \dots x_k^{r_k}$, where the summation extends over all $(r_1, r_2, \dots, r_k)$ such that $r_i \in \mathbb{N}$, $0 \leq r_i \leq n$, $\sum_{i=1}^k r_i = n$

i) WTS $\mathcal{P}(1)$ is true.

Left Side: $(x_1 + x_2 + \dots + x_k)^1 = x_1 + x_2 + \dots + x_k$

$0 \leq r_i \leq 1$, $\sum_{i=1}^k r_i = 1$

Since $r_i \in \mathbb{N}$, and $\sum_{i=1}^k r_i$, we know that exactly one $r_i = 1$ in any $(r_1, r_2, \dots, r_k)$.

Right Side: $\sum \frac{1!}{r_1!r_2!\dots r_k!} x_1^{r_1} x_2^{r_2} \dots x_k^{r_k} = \frac{1!}{1!0!\dots 0!} x_1^1 x_2^0 \dots x_k^0 + \frac{1!}{0!1!\dots 0!} x_1^0 x_2^1 \dots x_k^0 + \dots + \frac{1!}{0!0!\dots 1!} x_1^0 x_2^0 \dots x_k^1 = x_1 + x_2 + \dots + x_k$

Since The LHS = RHS, $\mathcal{P}(1)$ is true.

ii) WTS that $\mathcal{P}(n)$ is true $\implies \mathcal{P}(n+1)$ is true.

Let $n \in \mathbb{N}$ be given. Suppose $\mathcal{P}(n)$ is true.

Thus,

$(x_1 + x_2 + \dots + x_k)^n = \sum_{\sum r_i = n} \frac{n!}{r_1!r_2!\dots r_k!} x_1^{r_1} x_2^{r_2} \dots x_k^{r_k}$, where the summation extends over all $(r_1, r_2, \dots, r_k)$ such that $r_i \in \mathbb{N}$, $0 \leq r_i \leq n$, $\sum_{i=1}^k r_i = n$

$$\begin{aligned} (x_1 + x_2 + \dots + x_k)^{n+1} &= (x_1 + x_2 + \dots + x_k)^n (x_1 + x_2 + \dots + x_k) \\ &= \sum_{\sum r_i = n} \frac{n!}{r_1!r_2!\dots r_k!} x_1^{r_1+1} x_2^{r_2} \dots x_k^{r_k} + \sum_{\sum r_i = n} \frac{n!}{r_1!r_2!\dots r_k!} x_1^{r_1} x_2^{r_2+1} \dots x_k^{r_k} + \dots \\ &\quad + \sum_{\sum r_i = n} \frac{n!}{r_1!r_2!\dots r_k!} x_1^{r_1} x_2^{r_2} \dots x_k^{r_k+1} \\ &= \sum_{\sum r_i = n+1} \frac{n!}{(r_1-1)!r_2!\dots r_k!} x_1^{r_1} x_2^{r_2} \dots x_k^{r_k} + \sum_{\sum r_i = n+1} \frac{n!}{r_1!(r_2-1)!\dots r_k!} x_1^{r_1} x_2^{r_2} \dots x_k^{r_k} + \dots + \\ &\quad \sum_{\sum r_i = n+1} \frac{n!}{(r_1)!r_2!\dots (r_k-1)!} x_1^{r_1} x_2^{r_2} \dots x_k^{r_k} \\ &= \sum_{\sum r_i = n+1} \frac{(n+1)!}{r_1!r_2!\dots r_k!} x_1^{r_1} x_2^{r_2} \dots x_k^{r_k} \end{aligned}$$

By the generalization of Pascal's Law.

Thus, $\mathcal{P}(n)$ is true $\implies \mathcal{P}(n+1)$. Thus, by induction:

$(x_1 + x_2 + \dots + x_k)^n = \sum \frac{n!}{r_1!r_2!\dots r_k!} x_1^{r_1} x_2^{r_2} \dots x_k^{r_k}$, where the summation extends over all $(r_1, r_2, \dots, r_k)$ such that $r_i \in \mathbb{N}$, $0 \leq r_i \leq n$, $\sum_{i=1}^k r_i = n$. □

4. A.11 Show that the set $A = \{r : r \in \mathbb{Q}, r^2 < 2\}$ is bounded above in $\mathbb{Q}$ but has no l.u.b. in $\mathbb{Q}$.

Ans:

Proof. Let $x \in A$ be given. Thus, $x^2 < 2$.

$4 \in \mathbb{Q}$, and $4^2 = 16 > 2$. Thus, $x < 4$.

By the definition of an upper bound, 4 is an upper bound of A . $\square$

Proof. Suppose to the contrary that $\sup(A) = M$. Thus, $x \in A \implies x \leq M$, and $K < \sup(A) \implies \exists x \in A \text{ s.t. } x \leq M$ and that if $K < M$, $\exists x^* \in A \text{ s.t. } K < x^*$.

We know that $M < \sqrt{2}$. $M \neq \sqrt{2}$, since $\sqrt{2}$ is irrational.

$M, \sqrt{2} \in \mathbb{R}$, so by a corollary of the AOE, there exists a $b \in \mathbb{Q}$ such that $M < b < \sqrt{2}$.

There is a contradiction here, because $b < \sqrt{2} \implies b^2 < 2 \implies b \in A$. But, if M is the supremum of A this cannot be the case.

Thus, A has no l.u.b. in $\mathbb{Q}$. $\square$

5. A.12 Show that for any two sequences $\{x_n\}_{n \geq 1}, \{y_n\}_{n \geq 1} \subset \mathbb{R}$,

$$\underline{\lim}_{n \rightarrow \infty} x_n + \underline{\lim}_{n \rightarrow \infty} y_n \leq \underline{\lim}_{n \rightarrow \infty} (x_n + y_n) \leq \overline{\lim}_{n \rightarrow \infty} (x_n + y_n) \leq \overline{\lim}_{n \rightarrow \infty} x_n + \overline{\lim}_{n \rightarrow \infty} y_n$$

Ans:

Proof. Based on the definitions of infimums and supremums:

$$\inf_{k \geq n} x_k + \inf_{k \geq n} y_k \leq x_n + y_n \leq \sup_{k \geq n} x_k + \sup_{k \geq n} y_k.$$

$$\text{Thus, } \underline{\lim}_{n \rightarrow \infty} x_n + \underline{\lim}_{n \rightarrow \infty} y_n \leq \overline{\lim}_{n \rightarrow \infty} x_n + \overline{\lim}_{n \rightarrow \infty} y_n$$

It is also clear to see that $\inf_{k \geq n} (x_k + y_k) \leq \sup_{k \geq n} (x_k + y_k)$, which implies that:

$$\lim_{n \rightarrow \infty} \inf_{k \geq n} (x_k + y_k) \leq \lim_{n \rightarrow \infty} \sup_{k \geq n} (x_k + y_k) \iff$$

$$\underline{\lim}_{n \rightarrow \infty} (x_n + y_n) \leq \overline{\lim}_{n \rightarrow \infty} (x_n + y_n)$$

Putting this all together, we get:

$$\underline{\lim}_{n \rightarrow \infty} x_n + \underline{\lim}_{n \rightarrow \infty} y_n \leq \underline{\lim}_{n \rightarrow \infty} (x_n + y_n) \leq \overline{\lim}_{n \rightarrow \infty} (x_n + y_n) \leq \overline{\lim}_{n \rightarrow \infty} x_n + \overline{\lim}_{n \rightarrow \infty} y_n$$

$\square$

6. A.17 Find the radius of convergence, ρ , for the powers series $A(x) \equiv \sum_{n=0}^{\infty} a_n x^n$ where

(a) $a_n = \frac{n}{n+1}, n \geq 0$.

- (b) $a_n = n^p$, $n \geq 0$, $p \in \mathbb{R}$.
(c) $a_n = \frac{1}{n!}$, $n \geq 0$ (where $0! = 1$).

Ans: Suppose we have a sequence B of positive numbers. If $\log(B_n) \rightarrow b$ as $n \rightarrow \infty$, then $B \rightarrow e^b$ as $n \rightarrow \infty$. This fact will be used in the rest of the problem.

Additionally, if $\lim_{n \rightarrow \infty} a_n$ exists, $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \sup a_n = \lim_{n \rightarrow \infty} \inf a_n$ if a_n is an sequence of real numbers. This fact will also be used in the next parts.

$$(a) \rho = \left(\lim_{n \rightarrow \infty} \sup |a_n|^{\frac{1}{n}} \right)^{-1} = \left(\lim_{n \rightarrow \infty} \sup \left| \frac{n}{n+1} \right|^{\frac{1}{n}} \right)^{-1} = \left(\lim_{n \rightarrow \infty} \sup \left(\frac{n}{n+1} \right)^{\frac{1}{n}} \right)^{-1}.$$

$$\lim_{n \rightarrow \infty} \log \left(\left(\frac{n}{n+1} \right)^{\frac{1}{n}} \right) = \lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\frac{n}{n+1} \right) = \lim_{n \rightarrow \infty} \frac{\log(n) - \log(n+1)}{n}$$

$$\text{By L'Hopital's rule,} = \lim_{n \rightarrow \infty} \frac{\frac{1}{n} - \frac{1}{n+1}}{1} = \lim_{n \rightarrow \infty} \frac{1}{n^2 + n} \text{ By L'H,} = 0$$

$$\text{Thus, } \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^{\frac{1}{n}} = e^0 = 1 \implies \lim_{n \rightarrow \infty} \sup \left(\frac{n}{n+1} \right)^{\frac{1}{n}} = 1$$

$$\text{Thus, } \rho = (1)^{-1} = 1$$

$$(b) \rho = \left(\lim_{n \rightarrow \infty} \sup |a_n|^{\frac{1}{n}} \right)^{-1} = \left(\lim_{n \rightarrow \infty} \sup |n^p|^{\frac{1}{n}} \right)^{-1} = \left(\lim_{n \rightarrow \infty} \sup (n^{\frac{p}{n}}) \right)^{-1}$$

$$\lim_{n \rightarrow \infty} \log(n^{p/n}) = \lim_{n \rightarrow \infty} \frac{p \log(n)}{n}$$

$$\text{By L'H,} = \lim_{n \rightarrow \infty} \frac{p}{n} = p \lim_{n \rightarrow \infty} \frac{1}{n} = 0.$$

$$\text{Thus, } \lim_{n \rightarrow \infty} (n^{p/n}) = e^0 = 1 \implies \lim_{n \rightarrow \infty} \sup (n^{p/n}) = 1.$$

$$\text{Thus, } \rho = 1^{-1} = 1$$

$$(c) \rho = \left(\lim_{n \rightarrow \infty} \sup |a_n|^{\frac{1}{n}} \right)^{-1} = \left(\lim_{n \rightarrow \infty} \sup \left| \frac{1}{n!} \right|^{\frac{1}{n}} \right)^{-1}$$

We can show that $\lim_{n \rightarrow \infty} \left(\frac{1}{n} \right)^{1/n} = 0$ using L'H:

$$\lim_{n \rightarrow \infty} \left(\frac{1}{n} \right)^{1/n} = \lim_{n \rightarrow \infty} \frac{1}{n^{1/n}}$$

$$\text{By L'H,} = \lim_{n \rightarrow \infty} \frac{0}{\frac{1}{n} n^{1-1/n}} = \lim_{n \rightarrow \infty} 0 = 0$$

We know that $\frac{1}{n} \geq \frac{1}{n!}$ when $n > 0$. (Since we are looking at the limit as $n \rightarrow \infty$, we do not need to concern ourselves with what happens when $n = 0$).

Thus, $\left(\frac{1}{n} \right)^{1/n} \geq \left(\frac{1}{n!} \right)^{1/n}$ when $n > 0$. It is clear to see that $\left(\frac{1}{n!} \right)^{1/n}$ cannot be less than 0, and since $\left(\frac{1}{n} \right)^{1/n} \geq \left(\frac{1}{n!} \right)^{1/n}$, we can assume that $\lim_{n \rightarrow \infty} \left(\frac{1}{n!} \right)^{1/n} = 0$.

cannot be greater than $\lim_{n \rightarrow \infty} \left(\frac{1}{n}\right)^{1/n}$.

Thus, $\lim_{n \rightarrow \infty} \left(\frac{1}{n!}\right)^{1/n} = 0$.

Thus, $\lim_{n \rightarrow \infty} \sup \left(\frac{1}{n!}\right)^{1/n} = 0 \implies \rho = (0)^{-1} = \infty$.

This means that the radius of convergence diverges to $+\infty$

7. A.18

- (a) Find the Taylor series at $a = 0$ for the function $f(x) = \frac{1}{1-x}$ on $I \equiv (-1, +1)$ and show that it converges to $f(x)$ on I .
- (b) Find the Taylor series of $1 + x + x^2$ in $I = (1, 3)$, centered at 2.
- (c) Let

$$f(x) = \begin{cases} e^{-1/x^2} & \text{if } |x| < 1, x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

- (i) Show that f is infinitely differentiable at 0 and compute $f^{(j)}$ for all $j \geq 1$
- (ii) Show that the Taylor series at $a = 0$ converges but not to f on $(-1, 1)$.

Ans:

(a) $f^{(n)}(x) = \frac{n!}{(1-x)^{n+1}}$
 $\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n = \sum_{n=0}^{\infty} \frac{1}{n!} \frac{n!}{(1-0)^{n+1}} (x-0)^n = \sum_{n=0}^{\infty} x^n$ This is a geometric series, and it converges when $x \in (-1, 1) = I$ to $\frac{1}{1-x}$.

(b) $f^{(0)}(1) = 1 + 1 + (1)^2 = 3$, $f^{(1)}(1) = 1 + 2(1) = 3$, $f^{(2)}(1) = 2$,
 $f^{(3)}(1) = 0$, $f^{(4)}(1) = 0, \dots$

$$\begin{aligned} \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n &= \frac{3}{0!} (x-1)^0 + \frac{3}{1!} (x-1)^1 + \frac{2}{2!} (x-1)^2 + \frac{0}{3!} (x-1)^3 + 0 + \dots \\ &= 3 + 3x - 1 + x^2 - 2x + 1 + 0 + 0 + \dots \\ &= x^2 - x + 3 \text{ for } x \in I = (1, 3) \end{aligned}$$

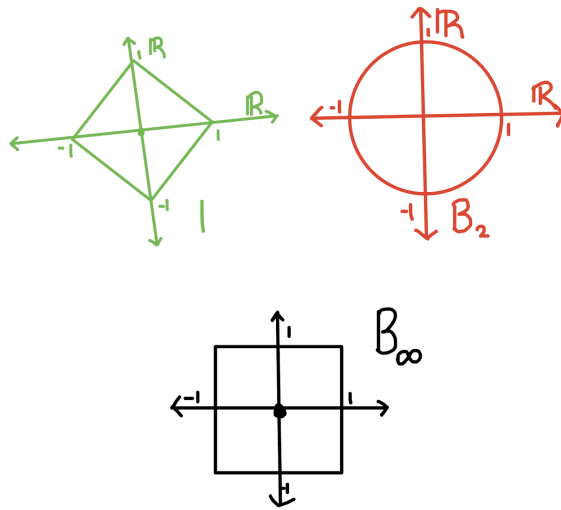
(c) (i) We know that $g(x) = e^{-x}$ is infinitely differentiable on x , as $g^{(j)}(x) = (-1)^j e^{-x}$ for all $j = 1, 2, \dots$, and we also know that $h(x) = \frac{1}{x^2} = x^{-2}$ is infinitely differentiable, as $h^{(j)}(x) = (-1)^j \frac{j!}{x^{2+j}}$.

By the chain rule, since both $g(x)$ and $h(x)$ are infinitely differentiable, we know that $f(x) = g(h(x))$ is also infinitely differentiable. And thus, $f^{(j)}(0) = 0$

(ii) $f^n = 0, \forall x \in \mathbb{N} \cup \{0\}$, so $\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} (x-0)^n = \sum_{n=0}^{\infty} 0 = 0$. Thus, regardless of your choice of $x \in (-1, 1) \cap \{0\}^c$, the Taylor series will converge to 0, not $f(x)$

8. A.24

(c) Draw the open unit ball $B_p \equiv \{x : x \in \mathbb{R}^2, d_p(x, 0) < 1\}$ in $\mathbb{R}^2$ for $p = 1, 2$ and ∞ .



The green is B_1 , the red is B_2 , and the black is B_∞